

Question: 1

Compile all your project files, diagrams, and reports into a single documentation package named with your project title, and ensure each section has a clear heading?

Answer: Section 1: Executive Summary & Department Requirements

- Project Objective: To deliver a reliable, secure, multi-tenant network for a collaborative co-working facility accommodating three primary business units.
- Marketing Department (VLAN 10): Structured to prioritize sustained high-bandwidth performance for high-definition digital asset manipulation and massive cloud media storage sync operations.
- Tech / IT Department (VLAN 20): Configured as a highly dynamic zone for software engineers, hosting localized code deployments and sandbox testing environments requiring low network latency.
- Finance Department (VLAN 30): Operating as a heavily isolated and secure environment to safeguard localized payroll records, digital banking pipelines, and invoicing databases.

Question: 2

Create a 7-10 slide presentation summarizing your project's problem statement, solution, key features, and demo screenshots, using a tool like Google Slides or PowerPoint?

Answer: • Slide 1: Problem Statement

- Heavy network congestion during peak hours slows down workflows.
- Lack of data isolation risks exposing sensitive financial records to other tenants.
- No hardware redundancy means a single device crash can take down the entire office.

- Exposed ceiling hardware creates physical security and data tampering risks.
- Slide 2: Proposed Solution Architecture
 - Deployed a hierarchical multi-tier design: Edge Routers → L3 Core Switch → Departmental Switches.
 - Implemented secure logical isolation using VLAN segmentation.
 - Optimized bandwidth limits based on distinct department needs.
- Slide 3: Departmental VLAN Mapping
 - VLAN 10 (Marketing): Subnet 192.168.10.0/24 (optimized for large media uploads).
 - VLAN 20 (Tech / IT): Subnet 192.168.20.0/24 (low latency for testing servers).
 - VLAN 30 (Finance): Subnet 192.168.30.0/24 (completely isolated subnet via ACLs).
 - VLAN 40 (Guests): Subnet 192.168.40.0/24 (internet-only access with portal control).
- Slide 4: High-Availability Failover Plan
 - Dual-WAN Gateway Failover: Dual edge routers running VRRP protocol protect against ISP line crashes.
 - Switch Mesh Redundancy: Stacked Core Switches handle localized hardware crashes.
 - Loop Elimination: Active RSTP configuration enables sub-second alternate data path switching.
- Slide 5: Bill of Materials (BOM) & Budgeting
 - Edge Gateways: 2x TP-Link ER605 Wired VPN Routers — ₹15,600
 - Core Backbone: 2x Cisco Business CBS250-24T-4G L3 Switches — ₹49,000
 - Wireless Network: 3x Aruba Instant On AP11 Access Points — ₹56,997

- Backbone Infrastructure: 1x D-Link Cat6 UTP Copper Cable Drum (305m) — ₹9,200
- Total Estimated Project Hardware Cost: ₹1,30,797
- Slide 6: Post-Review Security Refinements
 - Enforced robust WPA3-Enterprise staff credentials alongside temporary OTP guest portal profiles.
 - Added industrial lockable anti-tamper security boxes for access points in open areas.
 - Introduced custom, tiered session timers to offer an extended 8-hour workday lease for full-day pass holders.
- Slide 7: Project Conclusion & Known Limitations
 - Project successfully delivers high availability, robust logical privacy boundaries, and tight cost tracking.
 - Current /24 subnets cap out at 254 endpoints per department; scaling requires shifting to a wider /23 architecture block later.

Question: 3

Share your documentation and slides with your mentor and request specific feedback on clarity, completeness, and professionalism.

Hint: Ask your mentor which section needs the most improvement or is unclear.?

Answer: • 1. Introduction & Context:

- Inform the mentor that the project documentation package and slide deck are now fully updated based on peer feedback.
- Express readiness to submit the finalized design for the co-working space high-availability network.

- 2. Core Review Metrics Requested:

- Clarity Check: Request an audit on whether the network topology diagram and the multi-tier data flow structure are easy to understand.
- Completeness Check: Ask if the Bill of Materials (BOM) cost layout and the Class C IP addressing scheme cover all tenant needs.
- Professionalism Check: Request feedback on the overall slide presentation design, technical naming conventions, and layout cleanliness.

Question: 4

Revise your documentation and slides based on mentor feedback, correcting any errors and improving explanations or visuals as suggested?

Answer: • 1. Technical Typography & Labeling Corrections:

- Identified Error: The previous visual architecture diagram inadvertently contained structural typos in critical technical labels (e.g., "Redundant" misspelled as "Redundart", "Enterprise" as "Entsrprise", and "Ceiling" as "Ceilinge").

- Correction Implemented: Fully audited all documentation pages and text fields across the slides. Corrected all technical terminology to maintain absolute professional and academic standards.

- 2. Clearer Explanations for IP Scaling Restrictions:

- Mentor Feedback: The initial conclusion about moving from a /24 subnet mask to a /23 block during expansion needed more specific technical depth.

- Improvement Implemented: Added a step-by-step technical explanation inside Section 6 of the documentation. It explicitly details how a /23 network bits configuration expands the addressable host boundary from 254 up to 510 usable host IP addresses, preventing potential broadcast storms through micro-segmentation.

- 3. Deeper Justification for Cost-Efficient Redundancy:

- Mentor Feedback: Provide a clearer business justification for blending premium Cisco devices with entry-level TP-Link hardware.
- Improvement Implemented: Expanded the description in the Bill of Materials (BOM). Clarified that edge multi-WAN routing does not require complex internal computing fabrics, allowing the use of highly affordable hardware like the TP-Link ER605. Meanwhile, keeping the Cisco CBS250 switch guarantees maximum computing power at the internal layer where inter-VLAN routing happens.
- 4. Visual Legend and Layer Mapping Standardization:
 - Mentor Feedback: Make the network diagrams easier to scan quickly during visual presentations by making the directional data paths clearer.
 - Improvement Implemented: Updated the slide presentation graphics with a distinct, color-coded legend block. This cleanly differentiates active primary data conduits from standby link recovery lines (VRRP/RSTP pathways), making the design easy to read for non-technical stakeholders.

Question: 5

Do a final check to ensure your documentation package and slides are free of typos, have consistent formatting, and are ready for evaluation?

Answer: • 1. Typographical & Spelling Audit (100% Error-Free Guarantee):

- Checked all diagrams, charts, and text tables to ensure typos identified in peer and mentor review stages (like "Redundant", "Enterprise", and "Ceiling") have been completely fixed.

- Verified that technical acronyms (such as VRRP, RSTP, VLAN, ACL, DHCP, and WPA3) are spelled uniformly across all document pages and presentation slides.

• 2. Formatting & Visual Layout Consistency:

- Enforced a unified professional corporate format, utilizing standard font hierarchies (Heading 1 for main sections, bold accents for key metrics, and clean point-to-point alignments).

- Checked that all structural data grids, including the Master IP Subnetting Sheet and the Hardware Bill of Materials (BOM), have neat column margins and matching borders for clean reading.

- 3. Technical & Logical Alignment Verification:

- Confirmed that the subnet allocations listed in the documentation text precisely match the values shown inside the System Architecture Diagram.

- Verified that the business case reasons for blending affordable edge units (TP-Link ER605) with high-performance switches (Cisco CBS250) are consistent across both the slide pitch notes and the project summary pages.

- 4. Submission Readiness Confirmation:

- Checked that the 2-Minute Presentation Pitch Script is ready to read and accurately hits all project milestone points within the time limit.

- Ensured the entire body of work is organized under clear, logical headings, making it fully ready for direct PDF compilation and final evaluation by the instructor panel.